MATH 101 Unit 4: Exponentials, Logarithms, and Systems

Practice 1 | Version 202610122020.43 | 30 minutes

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right. Basic four-function calculator only.

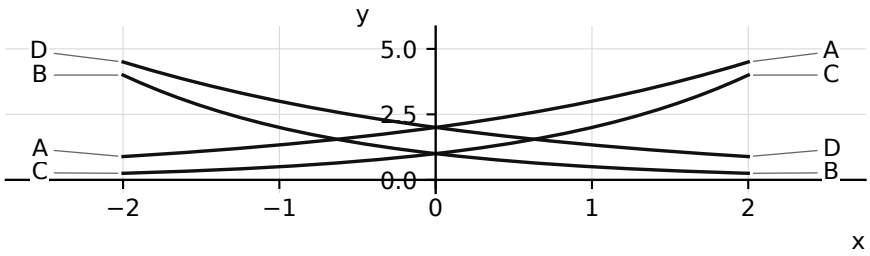
QUESTIONS	ANSWERS
1. Evaluate exactly using logarithmic properties; show the property used. Here log means base 10. (a) $\log(0.2) + \log(500)$ (b) $\frac{\log_3(8)}{\log_3(4)}$	1. (a) 10 pts (b) 10 pts
2. Solve both equations exactly. Check the logarithmic domain in (b). (a) $4^{x-1} = 2^{x+2}$ (b) $\log_2(x-1) + \log_2(3) = 2$	2. (a) 8 pts (b), with domain 12 pts
3. Find (a) domain and range; (b) vertical asymptote and x-intercept; (c) behavior at both ends of the domain. $f(x) = \log_2(x-1) + 1$	3. (a) Domain 4 pts (a) Range 4 pts (b) Asymptote 3 pts (b) x-intercept 3 pts (c) Left end 3 pts (c) Right end 3 pts

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QUESTIONS	ANSWERS
4. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 1(2)^x$ $(b)\quad 2(\frac{3}{2})^x$ $(c)\quad 2(\frac{2}{3})^x$ $(d)\quad 1(\frac{1}{2})^x$ 	4. (a) 5 pts (b) 5 pts (c) 5 pts (d) 5 pts
5. (a) Solve the linear system. (b) Solve the nonlinear system algebraically. (c) Test whether (0,0) belongs to the region given by the inequalities; show both comparisons. $(a)\quad x+y=3$ $x-y=1$ $(b)\quad x^2+y^2=5$ $y=x+1$ $(c)\quad x^2+y^2\le 5,\ y>x+1$	5. (a) Ordered pair 6 pts (b) Intersections 10 pts (c) Membership 4 pts

Instructor answer key: MATH 101 Unit 4: Exponentials, Logarithms, and Systems

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. (a)

2

10 pts

(b)

$\frac{3}{2}$

10 pts

2. (a)

$x = 4$

8 pts

(b), with domain

$x = \frac{7}{3};\ x > 1$

12 pts

3. (a) Domain

$(1, \infty)$

4 pts

(a) Range

$(-\infty, \infty)$

4 pts

(b) Asymptote

$x = 1$

3 pts

(b) x-intercept

$(\frac{3}{2},\ 0)$

3 pts

(c) Left end

$x \to 1^+ :\ f(x) \to -\infty$

3 pts

(c) Right end

$x \to \infty :\ f(x) \to \infty$

3 pts

TEST PAGE 2

4. (a)

C

5 pts

(b)

A

5 pts

(c)

D

5 pts

(d)

B

5 pts

5. (a) Ordered pair

$(2,\ 1)$

6 pts

(b) Intersections

$(1,\ 2);\ (-2,\ -1)$

10 pts

(c) Membership

At (0,0): 0 <= 5, but 0 > 1 is false; not in the region.

4 pts